

Evaluating reasoning large language models with human-like thinking in ophthalmic question answering

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ABSTRACT

Objectives To evaluate the performance of reasoning large language models (LLMs) with human-like thinking in ophthalmic question answering.

Methods We evaluated two state-of-the-art open-source reasoning LLMs (DeepSeek-R1 and QwQ-32B) and one conventional non-reasoning LLM (LLaMA-3.3-70B-Instruct) models on ophthalmology questions, assessing not only answer accuracy (ACC) but also the quality of their reasoning processes. First, we curated MedQA-Eye, a dataset of 967 ophthalmology questions across 10 subspecialties, 3 scenarios, 5 medical entities and 3 languages. Second, we proposed a novel framework considering human thinking patterns essential to medical practice to evaluate the thinking performance of reasoning LLMs on MedQA-Eye.

Results DeepSeek-R1 demonstrated superior overall ACC (90.59%, 95% CI 88.59% to 92.27%) to LLaMA-3.3-70B-Instruct (87.90%, 95% CI 85.69% to 89.81%, $p=0.015$) and QwQ-32B (84.28%, 95% CI 81.85% to 86.44%, $p<0.001$) with performance varying across subspecialties. Analysis of reasoning LLMs revealed incorrect logical inference as the primary point of failure, accounting for 93.41%–94.74% of incorrectly answered questions. We further quantified semantic uncertainty in reasoning LLM thinking as a predictor of answer reliability. DeepSeek-R1 exhibited lower semantic uncertainty (1.04 ± 3.63) compared with QwQ-32B (4.31 ± 40.70), $p<0.001$.

Conclusion Reasoning LLMs demonstrated superior performance in ophthalmology question answering, with DeepSeek-R1 achieving the highest ACC. Our findings demonstrate that reasoning LLM can better simulate human-like thinking processes compared with conventional non-reasoning LLM, suggesting its potential for more trustworthy LLM systems in ophthalmology.

INTRODUCTION

Large language models (LLMs) have demonstrated promise in medical question answering.^{1–4} Equipped with large-scale training data, LLM-based applications can play the role of assistant or copilot for clinicians.^{5–7} Furthermore, LLMs have been used for other purposes such as medical data extraction, structuring and analysis.^{8,9} Currently, the performance of LLMs in

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Reasoning large language models (LLMs) are developed to simulate human thinking processes through step-by-step logical reasoning, yet research on their performance in ophthalmic question answering remains limited.

WHAT THIS STUDY ADDS

⇒ This study curated an ophthalmic question-answering dataset (MedQA-Eye) and proposed a novel evaluation framework that aligns with human doctor thinking patterns to assess the performance of reasoning LLMs.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ By evaluating thinking performance, conclusion-oriented performance and semantic uncertainty of reasoning LLMs' thinking processes, our findings emphasise the need for more trustworthy LLM systems in ophthalmology applications. This work could guide future research focused on evaluating the human-like attributes of LLM systems in medical contexts.

medical knowledge has reached a new level, with experts preferring artificial intelligence (AI)-generated answers over those provided by generalist physicians.¹⁰ However, AI with 'black box' inherency is hardly to be trusted, especially when interacting with patients, unless it can adequately simulate the thinking processes of human experts.

The thinking processes of skilled human experts responding to queries typically involve three logical chains: (1) information collection: comprehending the question and context, (2) memory retrieval: recalling knowledge not provided in the context and (3) logical inference: performing non-trivial reasoning steps.¹¹ Although lots of research has tried strategies such as prompt engineering, retrieval augmented generation and agents' cooperation to improve the performance of LLMs on medical knowledge and to 'talk' more like a doctor, the reasoning

processes behind the models' outputs remain opaque to the user.^{9 12–14}

Recently, reasoning LLMs have garnered attention for their ability to solve complex problems through logical, structured and sequential reasoning processes.^{15 16} With data containing the chain-of-thought processes, reasoning LLMs were trained to 'think deeply' before answering a question. The 'thinking' processes include decomposing complex problems into a series of step-by-step logical subtasks, conducting self-reflection and systematically indexing relevant information. The goal of the reasoning LLM is to simulate human thinking and responding when faced with problems from the real world.

Ophthalmic diseases present with diverse manifestations and require prompt intervention. To investigate the potential application of reasoning LLMs in addressing ophthalmic questions, we evaluated the performance of two open-source reasoning LLMs and one conventional non-reasoning LLM. For the reasoning LLMs, we assessed not only answer accuracy (ACC) but also the quality of their reasoning processes to determine how effectively these models simulate the clinical thinking process of human doctors. Additionally, we analysed the semantic uncertainty in the reasoning LLMs' reasoning processes as a predictor of answer reliability.

MATERIALS AND METHODS

Dataset

The MedQA dataset is a standard benchmark for evaluating LLMs in medicine, which includes multiple choice questions validated by experts.¹⁷ The dataset involves each question's reference answer with an index, while the explanations were not included. To evaluate the performance of reasoning LLMs on ophthalmic diseases, we used questions concerning ophthalmology from the MedQA to construct MedQA-Eye. Three clinical experts annotated the questions of MedQA-Eye in several dimensions: (1) Classifying ophthalmic questions into a subspecialty of ophthalmology, referring to the hierarchy of the latest ICD-11.¹⁸ (2) Categorising ophthalmic questions into a topic scenario of medical practice (such as clinical or preclinical). (3) Labelling ophthalmic questions into a medical entity referring to the Unified Medical Language System.¹⁹ In the event of conflicting annotation conclusions, a committee discussion would be held to reach a consensus annotation conclusion (figure 1a). The language of each question was automatically recognised using Python's language detection libraries based on character encoding and linguistic patterns. Traditional Chinese was identified by traditional character forms, Simplified Chinese by simplified character forms and English by Latin alphabets. Given the distinct and non-overlapping character systems, the automated classification achieved 100% ACC, which was verified by clinical experts during the annotation process.

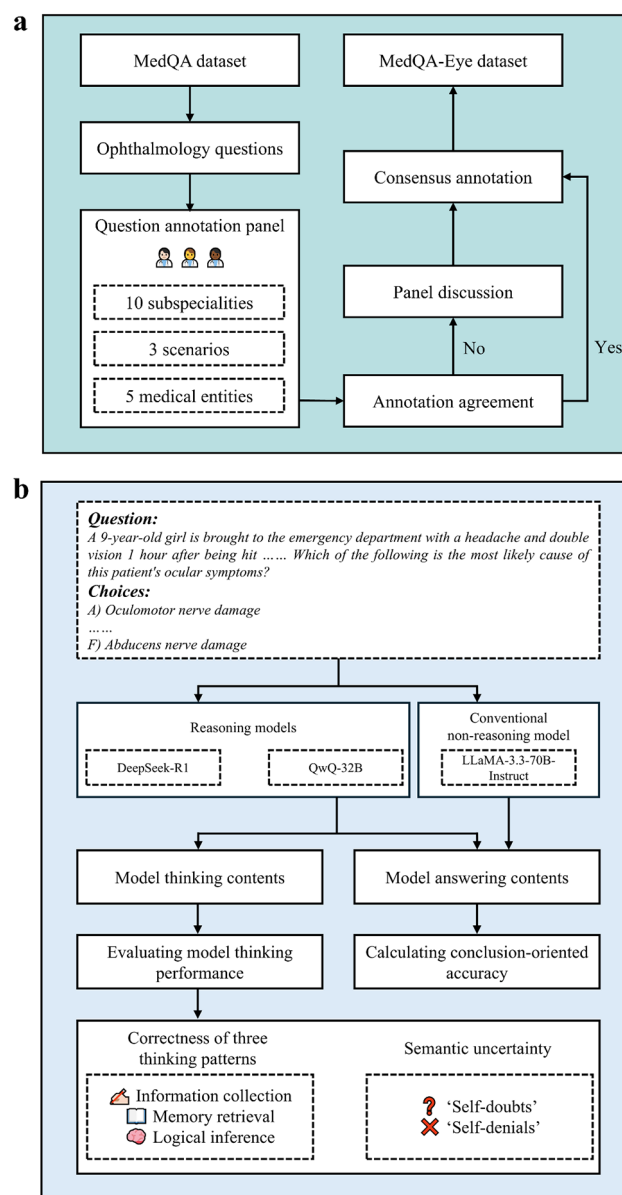


Figure 1 Construction process of the MedQA-Eye dataset and study workflow. (a) The construction process of MedQA-Eye began with selecting ophthalmology questions from the original MedQA dataset. These questions were annotated by three experts who classified each question according to three dimensions: subspecialty (based on ICD-11), scenario and medical entity (based on Unified Medical Language System). When experts' opinions conflicted, a committee discussion was held to reach consensus, resulting in the final MedQA-Eye dataset. (b) Questions and choices from MedQA-Eye were input into models to generate both thinking and answering contents. A multidisciplinary expert committee assessed thinking contents by labelling three thinking patterns as 'correct' or 'incorrect'. Semantic uncertainty was quantified by counting instances of 'self-doubts' or 'self-denials', providing insights into response reliability beyond traditional conclusion-oriented accuracy metrics. The answering contents of the models were recorded to calculate the conclusion-oriented accuracies.

Models

Although commercial LLMs (such as o1 by OpenAI and Claude-3.7-thinking by Anthropic) can generate reasoning contents, we focused on open-source models to make the study more repeatable and potentially applied by the broader scientific community. Therefore, we evaluated two state-of-the-art reasoning LLMs in the open-source community, DeepSeek-R1 (DeepSeek) from deepseek-ai²⁰ and QwQ-32B (QwQ) from Qwen.²¹ Both models were trained on data that contained ‘thinking’ contents built with chain-of-thought and self-reflection strategies, making the models ‘think’ deeply before answering a query. Moreover, the conventional non-reasoning LLM, LLaMA-3.3-70B-Instruct (LLaMA) from meta-llama,²² was also evaluated. All models were deployed and benchmarked in original precision without quantification to avoid potential performance reduction. All models were deployed on servers equipped with NVIDIA A40 *4 graphics processing units (GPUs) with central processing units (CPUs) offloads (NVIDIA Corporation, Santa Clara, California, USA).

Evaluating model thinking performance

Current LLM evaluation metrics rely predominantly on automated methods, such as bilingual evaluation under-study, but these approaches fail to adequately capture the complexity and nuances inherent in medical reasoning tasks.²³ Moreover, conclusion-oriented benchmarks (especially the datasets based on multiple choice questions) were widely used to estimate the performance of LLM, ignoring the rationality of their reasoning processes, which is not negligible in medical applications.²⁴ To evaluate the reasoning LLMs’ thinking performance on solving medical questions, we proposed a framework based on three thinking patterns that were necessary to a human doctor responding to a query: (1) information collection: comprehending the question and context, (2) memory retrieval: recalling knowledge not provided in the context and (3) logical inference: performing non-trivial reasoning steps. Experts independently analysed the thinking contents of DeepSeek and QwQ and dichotomously labelled the three thinking patterns as ‘correct’ or ‘incorrect’. For each question, three experts from the relevant subspecialty committee evaluated the thinking patterns, and the final label was determined by majority voting. For each thinking pattern, only the contents that were totally correct would be labelled as ‘correct’. Each of the three thinking patterns was evaluated independently, allowing for scenarios where a model could demonstrate correct performance in one or two patterns while failing in others. For instance, a response could exhibit correct information collection and memory retrieval but incorrect logical inference, with each pattern receiving its respective dichotomous label. This independent evaluation approach enabled us to identify the specific stages at which reasoning failures occurred during the question-answering process. For the authority of the subsequent subgroup analysis, we presented a study committee

composed of multidisciplinary experts to analyse the reasoning LLMs’ thinking contents. Preclinical medicine topics were analysed by laboratory scientists; medical teaching examination topics were analysed by professors from teaching hospitals; and the topics of drugs were analysed by the clinical pharmacists (figure 1, online supplemental figure S1).

Semantic uncertainty

Hallucination has been widely recognised as a significant and inevitable challenge for LLMs, as the possibility of generating content that appears correct but is incorrect always exists.²⁵ This challenge prevents users from trusting LLMs, even if the answers might be reasonable. Since non-expert users do not have the expertise to discern the generated answers, it is essential that the LLM system responds along with the reliability of their responses.²⁶ To quantify the reliability of reasoning LLMs’ responses, we defined the metrics of semantic uncertainty as ‘the number of self-doubts or self-denials that occur in model thinking’ (online supplemental figure S1). Self-doubts refer to instances where the model questions its reasoning, while self-denials involve explicit rejection of previous reasoning. Specific examples are provided in online supplemental table S1. In this study, we counted the semantic uncertainty for each tested question from the thinking processes of DeepSeek and QwQ to evaluate how uncertain the models were for their responses.

Statistical analysis

To report the performance of DeepSeek, QwQ and LLaMA, we calculated the ACC of the three models on the MedQA-Eye dataset. CI was calculated using the Wilson Score Interval.²⁷ The differences between the models and subgroups were analysed using the McNemar test. The differences in the semantic uncertainty between the reasoning LLMs were analysed using the Wilcoxon signed-rank test. Statistical analysis was performed using Python V.3.10.14 (Wilmington, Delaware, USA), with statistical significance set at a two-sided p value below 0.05 for individual tests and below 0.0167 after Bonferroni correction for multiple comparisons. Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

RESULTS

Characteristics of the MedQA-Eye dataset

After the curation of MedQA-Eye, 967 questions across 10 ophthalmic subspecialties, 3 scenarios of medical practices, 5 medical entities and 3 languages were enrolled. The subspecialties included anterior segment, visual pathways or centres, posterior segment, ocular adnexa or orbit, strabismus or ocular motility disorders, glaucoma, refraction of accommodation, ocular tumour, ocular trauma and others. The scenarios of medical practices included clinical medicine, preclinical medicine and medical teaching examination. The medical entities included diagnosis, drug, anatomy, surgery and others.

Table 1 Distributions of the MedQA dataset

	Number	Ratio
Subspeciality		
Anterior segment	286	29.58%
Visual pathways or centres	158	16.34%
Posterior segment	150	15.51%
Ocular adnexa or orbit	97	10.03%
Strabismus or ocular motility disorders	95	9.82%
Glaucoma	87	9.00%
Refraction of accommodation	38	3.93%
Ocular tumour	27	2.79%
Ocular trauma	19	1.96%
Others	10	1.03%
Scenario		
Medical teaching examination	783	80.97%
Clinical medicine	171	17.68%
Preclinical medicine	13	1.34%
Medical entity		
Diagnosis	741	76.63%
Drug	96	9.93%
Anatomy	61	6.31%
Surgery	40	4.14%
Others	29	3.00%
Language		
Traditional Chinese	452	46.74%
English	334	34.54%
Simplified Chinese	181	18.72%

The languages included Traditional Chinese, English and Simplified Chinese. The details of the MedQA-Eye dataset are arranged in [table 1](#).

Conclusion-oriented performance of three LLMs

We evaluated three open-source LLMs on the MedQA-Eye dataset. DeepSeek demonstrated superior overall ACC at 90.59% (95% CI 88.59% to 92.27%), followed by LLaMA at 87.90% (95% CI 85.69% to 89.81%, $p=0.015$) and QwQ at 84.28% (95% CI 81.85% to 86.44%, $p<0.001$). In subspecialty evaluation ([table 2](#) and [figure 2](#)), DeepSeek significantly outperformed QwQ in several key areas, including anterior segment questions ($p<0.001$), visual pathways or centres ($p=0.009$) and ocular adnexa or orbit ($p=0.016$). Although LLaMA is a conventional non-reasoning LLM, it significantly outperformed the reasoning LLM QwQ in visual pathways or centres ($p=0.037$) and ocular adnexa or orbit ($p=0.024$). Moreover, LLaMA performed comparably to DeepSeek across most subspecialties, with the only statistically significant difference observed in refraction of accommodation questions ($p=0.041$), where DeepSeek demonstrated

superior performance. In evaluating performance across three medical practice scenarios ([figure 3a](#)), both DeepSeek ($p<0.001$) and LLaMA ($p=0.023$) significantly outperformed QwQ in addressing medical teaching examination questions, with DeepSeek also demonstrating an advantage over LLaMA ($p=0.036$). In clinical medicine questions, DeepSeek performed superior to QwQ ($p=0.008$), while LLaMA and DeepSeek performed with comparable effectiveness. In preclinical medicine questions, all three models performed similarly, though the small sample size ($n=13$) limits the statistical power of this comparison. In evaluation from the dimension of medical entities ([figure 3b](#)), both DeepSeek and LLaMA significantly outperformed QwQ ($p<0.001$ and $p=0.002$, respectively) in questions regarding diagnosis, while showing comparable performance to each other. The three LLMs demonstrated comparable performance for drug, anatomy and surgery questions. For questions of the 'others' category, DeepSeek significantly outperformed LLaMA ($p<0.001$), with no other significant differences observed. In language performance evaluation ([figure 3c](#)), DeepSeek significantly outperformed QwQ in both Traditional Chinese and English ($p<0.001$ for both), while LLaMA exceeded QwQ in English ($p=0.004$). DeepSeek performed significantly better than LLaMA in Simplified Chinese ($p=0.009$).

Reasoning LLMs thinking performance

Three thinking patterns that were necessary to a human doctor responding to a query were labelled 'correct' or 'incorrect'. An example of assessing the reasoning LLM thinking process is presented in online supplemental figure S1. We reported the frequencies of occurrence for the three thinking patterns in online supplemental table S2. Our analysis revealed striking contrasts in thinking processes between correct and incorrect responses. For DeepSeek, in correctly answered questions, we observed near-perfect rates of correct information collection (99.20%), accurate memory retrieval (98.86%) and sound logical inference (97.72%). For QwQ, correctly answered questions similarly demonstrated high rates of correct information collection (97.06%), memory retrieval (96.20%) and logical inference (92.39%). Conversely, responses leading to incorrect answers exhibited substantial deficiencies across all three thinking patterns in both models. Notably, while approximately 30.77% of DeepSeek's incorrect responses and 42.76% of QwQ's incorrect responses still demonstrated proper information collection, and about 30.77% (DeepSeek) and 19.08% (QwQ) maintained accurate memory retrieval, merely 6.59% (DeepSeek) and 5.26% (QwQ) exhibited correct logical inference. This suggests that the critical failure point in incorrect responses predominantly occurs at the logical inference stage, rather than during information gathering or recall. These findings highlight that even when foundational processes of information collection and memory retrieval function

Table 2 Comparison of three models' conclusion-oriented performances on 10 subspecialties of ophthalmology, 3 scenarios of medical practices, 5 medical entities and 3 languages

	DeepSeek ACC (95% CI)	QwQ ACC (95% CI)	LLaMA ACC (95% CI)	P1	P2	P3
Subspeciality						
Anterior segment	91.96% (88.22% to 94.58%)	85.11% (80.48% to 88.79%)	88.46% (84.24% to 91.67%)	0.089	<0.001	0.155
Visual pathways or centres	91.14% (85.68% to 94.65%)	85.44% (79.10% to 90.10%)	91.14% (85.68% to 94.65%)	0.803	0.009	0.037
Posterior segment	84.67% (78.04% to 89.56%)	80.65% (73.72% to 86.09%)	84.67% (78.04% to 89.56%)	0.789	0.332	0.332
Ocular adnexa or orbit	88.66% (80.83% to 93.55%)	78.35% (69.16% to 85.38%)	88.66% (80.83% to 93.55%)	0.789	0.016	0.024
Strabismus or ocular motility disorders	88.42% (80.45% to 93.41%)	82.80% (73.87% to 89.12%)	87.37% (79.21% to 92.62%)	1.000	0.181	0.383
Glaucoma	94.25% (87.24% to 97.52%)	87.50% (78.99% to 92.87%)	87.36% (78.76% to 92.79%)	0.114	0.077	0.803
Refraction of accommodation	97.37% (86.51% to 99.53%)	94.44% (81.86% to 98.46%)	81.58% (66.58% to 90.78%)	0.041	1.000	0.182
Ocular tumour	96.30% (81.72% to 99.34%)	92.59% (76.63% to 97.94%)	100.00% (87.54% to 100.00%)	1.000	0.480	1.000
Ocular trauma	94.74% (75.36% to 99.06%)	84.21% (62.43% to 94.48%)	84.21% (62.43% to 94.48%)	0.500	1.000	1.000
Others	90.00% (59.58% to 98.21%)	83.33% (55.20% to 95.30%)	70.00% (39.68% to 89.22%)	0.500	1.000	1.000
Scenario						
Medical teaching examination	91.06% (88.86% to 92.86%)	85.19% (82.53% to 87.50%)	88.51% (86.08% to 90.55%)	0.036	<0.001	0.023
Clinical medicine	87.72% (81.96% to 91.83%)	79.53% (72.87% to 84.90%)	84.21% (78.00% to 88.92%)	0.307	0.008	0.186
Preclinical medicine	100.00% (77.19% to 100.00%)	92.31% (66.69% to 98.63%)	100.00% (77.19% to 100.00%)	1.000	1.000	1.000
Medical entity						
Diagnosis	89.47% (87.06% to 91.48%)	83.00% (80.12% to 85.53%)	87.45% (84.87% to 89.64%)	0.129	<0.001	0.002
Drug	93.75% (87.03% to 97.10%)	92.71% (85.71% to 96.42%)	89.58% (81.88% to 94.24%)	0.221	1.000	0.450
Anatomy	97.50% (87.12% to 99.56%)	90.00% (76.95% to 96.04%)	85.00% (70.93% to 92.94%)	0.752	0.131	0.579
Surgery	91.80% (82.21% to 96.45%)	83.61% (72.39% to 90.84%)	88.52% (78.16% to 94.33%)	0.074	0.371	0.752
Others	96.55% (82.82% to 99.39%)	82.76% (65.45% to 92.40%)	96.55% (82.82% to 99.39%)	<0.001	0.134	0.134
Language						
Traditional Chinese	90.49% (87.43% to 92.86%)	84.96% (81.37% to 87.96%)	88.50% (85.22% to 91.12%)	0.253	<0.001	0.069
English	87.72% (83.77% to 90.82%)	79.64% (75.00% to 83.61%)	86.53% (82.45% to 89.78%)	0.617	<0.001	0.004
Simplified Chinese	96.13% (92.23% to 98.11%)	91.16% (86.12% to 94.49%)	88.95% (83.55% to 92.73%)	0.009	0.052	0.540

P1 refers to the p value calculated between DeepSeek and LLaMA using the two-sided McNemar test. P2 refers to the p value calculated between the DeepSeek and QwQ using the two-sided McNemar test. P3 refers to the p value calculated between the QwQ and LLaMA using the two-sided McNemar test.

Bold values indicate statistically significant differences ($p < 0.05$).

ACC, accuracy; DeepSeek, DeepSeek-R1; LLaMA, LLaMA-3.3-70B-Instruct; QwQ, QwQ-32B.

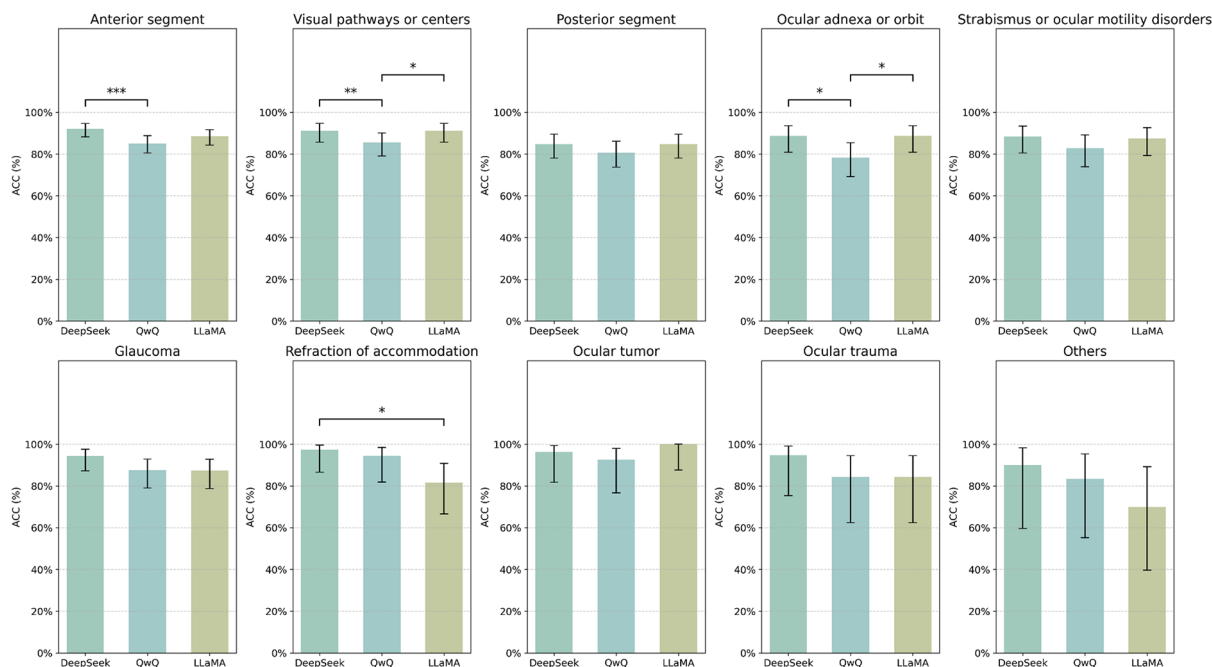


Figure 2 Performances of three models across 10 subspecialties of ophthalmology. Bar plots show the ACC (%) with 95% CI for DeepSeek, QwQ and LLaMA on the MedQA-Eye across anterior segments, visual pathways or centres, posterior segment, ocular adnexa or orbit, strabismus or ocular motility disorders, glaucoma, refraction of accommodation, ocular tumour, ocular trauma and others. Statistical significances between models are indicated by asterisks (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$). DeepSeek generally demonstrates higher ACC across most subspecialties, with statistically significant differences observed in the anterior segment, visual pathways or centres, ocular adnexa or orbit and refraction of accommodation categories. ACC, accuracy; DeepSeek, DeepSeek-R1; LLaMA, LLaMA-3.3-70B-Instruct; QwQ, QwQ-32B.

adequately, flawed logical inference almost invariably leads to erroneous conclusions. This indicates that logical inference deficits represent the most significant barrier to accurate problem-solving in both models.

Semantic uncertainty as a predictor of model performance

We quantified semantic uncertainty in two advanced reasoning LLMs when applied to ophthalmic questions (online supplemental figure S2). DeepSeek demonstrated substantially lower overall semantic uncertainty (1.04 ± 3.63) than QwQ (4.31 ± 40.70) across the MedQA-Eye dataset. This pattern persisted when stratifying by answer correctness. For correctly answered questions, DeepSeek maintained remarkably low semantic uncertainty (0.75 ± 3.15) while QwQ exhibited higher variability (4.09 ± 44.16). Notably, both models showed elevated semantic uncertainty when producing incorrect answers, with DeepSeek increasing approximately five-fold (3.81 ± 5.95) and QwQ showing more moderate elevation (5.44 ± 8.94). These findings establish semantic uncertainty as a potential quantitative indicator of reasoning reliability in medical contexts. The pronounced difference between correct and incorrect responses, particularly in DeepSeek, suggests that semantic uncertainty metrics could serve as effective warning signals for potentially erroneous model outputs, providing valuable guardrails for clinical applications.

DISCUSSION

This study represents a comprehensive evaluation of reasoning LLMs' performance on ophthalmic question answering, focusing not only on output ACC but also on the quality of the underlying reasoning processes. We evaluated two state-of-the-art open-source reasoning LLMs (DeepSeek and QwQ) alongside a conventional non-reasoning LLM (LLaMA) on the MedQA-Eye dataset, which we curated to encompass 967 questions across diverse ophthalmological subspecialties, scenarios of medical practices, medical entities and languages. Our evaluation extended beyond traditional ACC metrics to analyse the models' thinking processes, examining their performance in information collection, memory retrieval and logical inference, mirroring the thinking patterns of human doctors.

Our findings revealed several notable trends with significant implications for LLM medical applications. While DeepSeek demonstrated superior overall ACC (90.59%, 95% CI 88.59% to 92.27%), all models showed substantial performance variations across ophthalmological subspecialties. Analysis of reasoning LLMs revealed incorrect logical inference as the primary point of failure, accounting for 93.41%–94.74% of incorrectly answered questions, despite 30.77% maintaining correct information collection and memory retrieval. Additionally, we identified semantic uncertainty as a potential quantitative indicator of reasoning reliability, observing

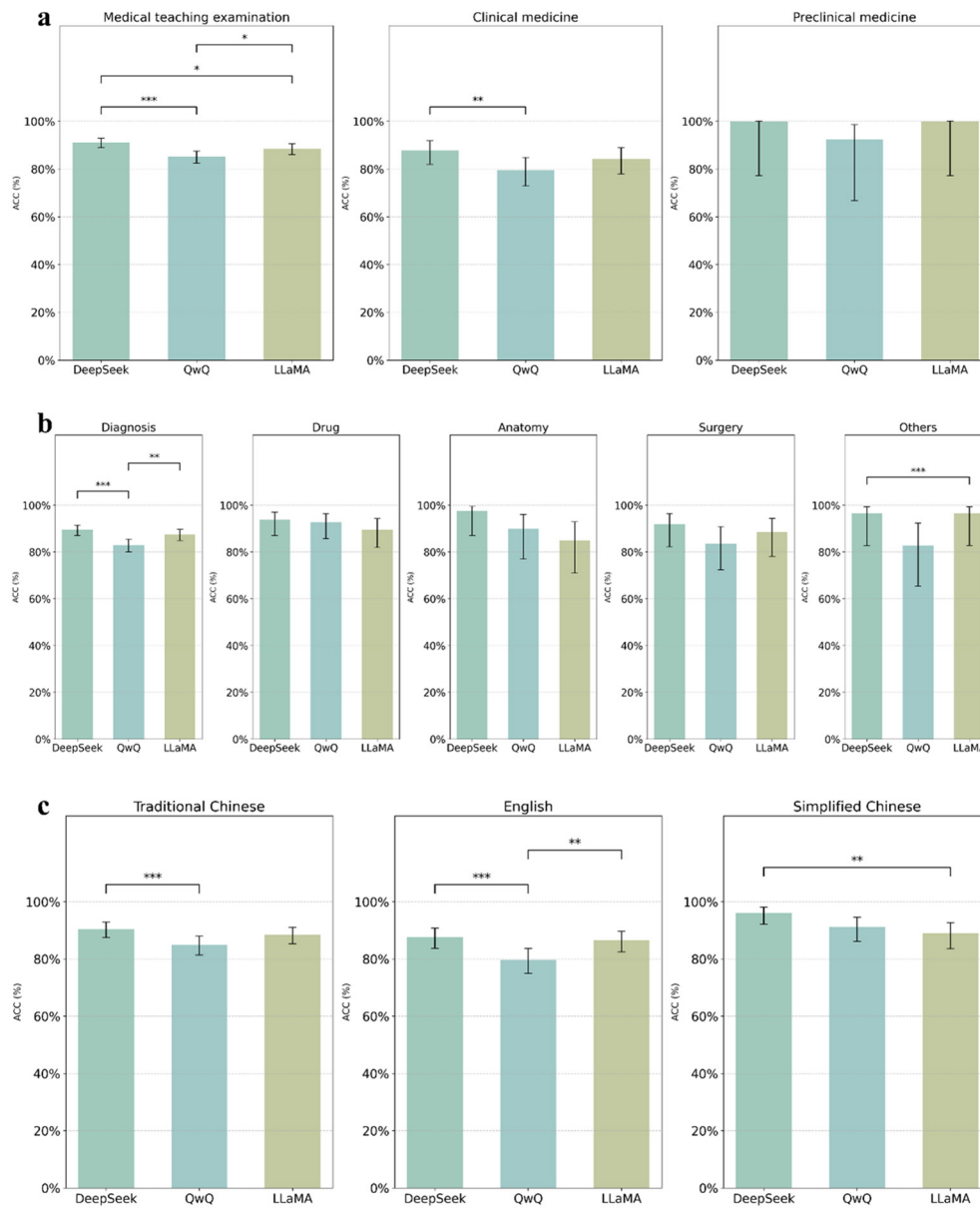


Figure 3 Performances of three models across scenarios, medical entities and languages. Bar plots show the ACC (%) with 95% CI for DeepSeek, QwQ and LLaMA on the MedQA-Eye. Statistical significances between models are indicated by asterisks (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$). (a) The models' performances across three scenarios of medical practices. (b) The models' performances across five medical entities. (c) The models' performances across three languages. ACC, accuracy; DeepSeek, DeepSeek-R1; LLaMA, LLaMA-3.3-70B-Instruct; QwQ, QwQ-32B.

that semantic uncertainty scores increased substantially when models generated incorrect answers. This effect was particularly pronounced in the DeepSeek model, where semantic uncertainty increased nearly fivefold for incorrect responses.

The cross-linguistic analysis revealed important differential performance patterns that merit further consideration. DeepSeek demonstrated consistently strong performance across all three languages (87.72%–96.13%), while QwQ showed notable weaknesses in Traditional Chinese (84.96%) and particularly in English (79.64%). QwQ's weaker English performance compared with its Chinese performance may be attributed to several factors: (1) training data

composition, where the model may have been exposed to a higher proportion of Chinese-language medical content, (2) language-specific challenges in medical reasoning, as medical terminology and logical structures may vary across languages and (3) tokenisation efficiency differences, which can affect the model's ability to process and reason in different languages. Interestingly, DeepSeek significantly outperformed LLaMA in Simplified Chinese (96.13% vs 88.95%, $p = 0.009$), suggesting potential advantages for localised medical applications. These language-specific variations highlight the importance of considering linguistic context and training data diversity when deploying AI systems across diverse healthcare environments.

Further analysis of subspecialty performance revealed notable variations in model capabilities across ophthalmological domains. All three models demonstrated exceptional ACC in ocular tumour questions (DeepSeek: 96.30%, QwQ: 92.59%, LLaMA: 100.00%), with LLaMA achieving perfect performance in this subspecialty. However, this subset contained only 27 questions, limiting the statistical power and generalisability of this finding. The exceptional performance of LLaMA in this domain may suggest that ocular tumour questions in our dataset primarily involve pattern recognition and factual recall rather than complex multi-step reasoning, which could favour conventional non-reasoning LLMs. Larger datasets are needed to validate these subspecialty-specific performance patterns. Higher ACC was observed for refraction of accommodation questions in the reasoning model, where DeepSeek (97.37%) significantly outperformed LLaMA (81.58%, $p=0.041$). Posterior segment disorders presented greater challenges across all models (DeepSeek: 84.67%, QwQ: 80.65%, LLaMA: 84.67%), representing one of the lowest-performing subspecialties in our evaluation. Although DeepSeek significantly outperformed QwQ in anterior segment questions (91.96% vs 85.11%, $p<0.001$), visual pathways or centres (91.14% vs 85.44%, $p=0.009$) and ocular adnexa or orbit (88.66% vs 78.35%, $p=0.016$), LLaMA, being a conventional non-reasoning LLM, performed comparably to DeepSeek in these areas. These subspecialty-specific performance variations highlight the potential specialised applications of different LLMs in ophthalmology, with reasoning LLMs demonstrating advantages in domains requiring complex inference, while suggesting that conventional non-reasoning LLMs may still offer competitive performance for specific categories.

Previous studies evaluating LLMs have primarily focused on conclusion-oriented metrics, overlooking the crucial aspect of reasoning quality in medical practices. Nori *et al* (Microsoft and OpenAI) evaluated medical knowledge in LLMs using traditional ACC benchmarks, while Marcos *et al* assessed LLMs' performance with retrieval-augmented functions on health questions without examining reasoning processes.^{28 29} Savage *et al* acknowledged the importance of model transparency and tried to mitigate the 'black box' limitations of LLMs by developing reasoning prompts, but lacked quantitative metrics for evaluating reasoning reliability.³⁰ Our study advances the field by introducing a structured framework for evaluating the reasoning process, alongside quantifiable metrics of semantic uncertainty that could serve as a built-in warning system for potentially erroneous outputs. These approaches can potentially enhance the trustworthiness of LLMs in medical practices, establishing a framework for more transparent and reliable AI systems in ophthalmology.

Our study has several limitations that should be acknowledged. First, while MedQA-Eye represents a diverse collection of ophthalmological questions, it may not fully capture the complexity and nuance of

real-world clinical scenarios, which often involve ambiguous presentations and incomplete information. Second, our evaluation focused on multiple choice questions, which, despite providing standardised assessment, may not adequately reflect the open-ended nature of many clinical queries where physicians must formulate differential diagnoses, weigh multiple possibilities and make decisions under uncertainty. This structured format may inadvertently favour pattern recognition over genuine clinical reasoning and could overestimate model performance compared with real-world applications. Third, although we analysed semantic uncertainty as a potential indicator of reasoning reliability, this metric requires further validation in prospective studies before clinical implementation. In future works, we would curate more extensive and diverse datasets incorporating open-ended diagnostic tasks that better simulate authentic clinical decision-making, potentially integrating multimodal data including smartphone-based imaging,³¹ conducting prospective evaluation in real-world clinical settings with actual patient cases, and evaluating semantic uncertainty as a clinical reliability indicator across diverse healthcare settings and patient populations.

CONCLUSION

In conclusion, we conducted a comprehensive evaluation of open-source reasoning LLMs in ophthalmic question answering, revealing that these models can achieve high ACC while demonstrating human-like thinking processes. Our novel analysis framework revealed insights into how these models 'think': logical inference was identified as the critical failure point in incorrect responses. We also established semantic uncertainty as a quantifiable predictor of answer reliability. These findings represent significant progress towards more transparent and trustworthy AI systems in ophthalmology. Future research should focus on enhancing logical inference capabilities and refining uncertainty metrics as guardrails for clinical AI, ultimately working toward reasoning LLMs that can serve as reliable, transparent assistants in ophthalmic care while maintaining the human-like thinking processes essential to medical decision-making.

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